
List of Publications

Publications (Peer-reviewed)

Monographs

Book Publications

1. A. Kaltenbach. *Pseudo-monotone operator theory for unsteady problems with variable exponents*, **Lecture Notes in Mathematics**, 2023.
DOI: [10.1007/978-3-031-29670-3](https://doi.org/10.1007/978-3-031-29670-3)
2. E. Emmrich and A. Kaltenbach. *Funktionen mit Werten in einem Banach-Raum: Funktionenräume für zeitabhängige Probleme*, **Springer-Lehrbuch**, to appear, 2026.
DOI: [10.1007/978-3-662-72999-1](https://doi.org/10.1007/978-3-662-72999-1)

PhD Thesis

3. A. Kaltenbach. *Theory of pseudo-monotone operators for unsteady problems in variable exponent spaces*, **Freidok University of Freiburg**, 2021.
DOI: [10.6094/UNIFR/222538](https://doi.org/10.6094/UNIFR/222538)

Journal Publications

4. L. C. Berselli and A. Kaltenbach. *Pulsatile Flows for Simplified Smart Fluids with Variable Power-Law: Analysis and Numerics*, submitted, **Mathematical Models and Methods in Applied Sciences**, accepted, 2026.
URL: <https://arxiv.org/abs/2507.22449>
5. A. Kaltenbach and J. Wichmann. *A Priori Error Analysis for the p -Stokes Equations with Slip Boundary Conditions: A Discrete Leray Projection Framework*, **IMA Journal of Numerical Analysis**, accepted, 2026.
URL: <https://arxiv.org/abs/2507.15016>
6. H. Antil, A. Kaltenbach, and K. Kirk. *Duality-based algorithm and numerical analysis for optimal insulation problems on non-smooth domains*, **SIAM Journal on Control and Optimization**, 2026.
DOI: [10.1137/25M1758647](https://doi.org/10.1137/25M1758647)
7. J. Blechta, P. A. Gazca-Orozco, A. Kaltenbach, and M. Růžička. *Quasi-optimal discontinuous Galerkin discretizations of the p -Dirichlet problem*, **Numerische Mathematik**, 2026.
DOI: [10.1007/s00211-025-01507-1](https://doi.org/10.1007/s00211-025-01507-1)
8. A. Kaltenbach and M. Zeinhofer. *The Deep Ritz Method for parametric p -Dirichlet problems*, **Advances in Continuous and Discrete Models: Theory and Applications**, 2025.
DOI: [10.1186/s13662-025-04043-2](https://doi.org/10.1186/s13662-025-04043-2)
9. H. Antil, A. Kaltenbach, and K. Kirk. *Mathematical modelling for an optimal insulation problem on Lipschitz domains*, **SIAM Journal on Mathematical Analysis**, 2025.
DOI: [10.1137/25M1741947](https://doi.org/10.1137/25M1741947)
10. H. Antil, S. Bartels, A. Kaltenbach, and R. Khandelwal. *Variational problems with gradient constraints: A priori and a posteriori error identities*, **Mathematics of Computation**, 2025.
DOI: [10.1090/mcom/4146](https://doi.org/10.1090/mcom/4146)
11. S. Bartels, T. Gudi, and A. Kaltenbach. *A priori and a posteriori error identities for the scalar Signorini problem*, **SIAM Journal on Numerical Analysis**, 2025.
DOI: [10.1137/24M1677691](https://doi.org/10.1137/24M1677691)
12. S. Bartels and A. Kaltenbach. *Error analysis for a Crouzeix-Raviart approximation of the obstacle problem*, **Journal of Numerical Mathematics**, 2025.
DOI: [10.1515/jnma-2025-0036](https://doi.org/10.1515/jnma-2025-0036)
13. L. C. Berselli and A. Kaltenbach. *Convergence analysis of a fully-discrete finite element approximation of the unsteady $p(\cdot, \cdot)$ -Navier-Stokes equations*, **Numerische Mathematik**, 2025.
DOI: [10.1007/s00211-025-01450-1](https://doi.org/10.1007/s00211-025-01450-1)

14. L. C. Berselli, A. Kaltenbach, and M. Růžička. *Energy conservation for weak solutions of incompressible Newtonian fluid equations in Hölder spaces with Dirichlet boundary conditions in the half-space*, **Mathematische Annalen**, 2024.
DOI: [10.1007/s00208-024-03065-7](https://doi.org/10.1007/s00208-024-03065-7)
15. L. C. Berselli and A. Kaltenbach. *Error analysis for a finite element approximation of the steady $p(\cdot)$ -Navier–Stokes equations*, **IMA Journal of Numerical Analysis**, 2024.
DOI: [10.1093/imanum/drae082](https://doi.org/10.1093/imanum/drae082)
16. S. Bartels and A. Kaltenbach. *Explicit a posteriori error representation for variational problems and application to TV-minimization*, **Foundations of Computational Mathematics**, 2024.
DOI: [10.1007/s10208-024-09676-5](https://doi.org/10.1007/s10208-024-09676-5)
17. A. Kaltenbach and M. Růžička. *Note on quasi-optimal error estimates for the pressure for shear-thickening fluids*, **ESAIM: Mathematical Modelling and Numerical Analysis**, 2024.
DOI: [10.1051/m2an/2024051](https://doi.org/10.1051/m2an/2024051)
18. J. Jeřberger and A. Kaltenbach. *Finite element discretization of the steady, generalized Navier–Stokes equations with inhomogeneous Dirichlet boundary conditions*, **SIAM Journal on Numerical Analysis**, 2024.
DOI: [10.1137/23M1607398](https://doi.org/10.1137/23M1607398)
19. A. Kh. Balci and A. Kaltenbach. *Error analysis for a Crouzeix–Raviart approximation of the variable exponent Dirichlet problem*, **IMA Journal of Numerical Analysis**, 2024.
DOI: [10.1093/imanum/drae025](https://doi.org/10.1093/imanum/drae025)
20. P. A. Gazca–Orozco and A. Kaltenbach. *On the stability and convergence of Discontinuous Galerkin schemes for incompressible flows*, **IMA Journal of Numerical Analysis**, 2024.
DOI: [10.1093/imanum/drae004](https://doi.org/10.1093/imanum/drae004)
21. A. Kaltenbach. *Error analysis for a Crouzeix–Raviart approximation of the p -Dirichlet problem*, **Journal of Numerical Mathematics**, 2023.
DOI: [10.1515/jnma-2022-0106](https://doi.org/10.1515/jnma-2022-0106)
22. A. Kaltenbach and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier–Stokes system, Part III: Convergence rates for the pressure*, **SIAM Journal on Numerical Analysis**, 2023.
DOI: [10.1137/22M1541472](https://doi.org/10.1137/22M1541472)
23. A. Kaltenbach and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier–Stokes system, Part II: Convergence rates for the velocity*, **SIAM Journal on Numerical Analysis**, 2023.
DOI: [10.1137/22M1514751](https://doi.org/10.1137/22M1514751)
24. A. Kaltenbach and M. Růžička. *A Local Discontinuous Galerkin approximation for the p -Navier–Stokes system, Part I: Convergence analysis*, **SIAM Journal on Numerical Analysis**, 2023.
DOI: [10.1137/22M151474X](https://doi.org/10.1137/22M151474X)
25. A. Kaltenbach and M. Růžička. *Convergence analysis of a Local Discontinuous Galerkin approximation for systems with Olicz–structure*, **ESAIM: Mathematical Modelling and Numerical Analysis**, 2023.
DOI: [10.1051/m2an/2023028](https://doi.org/10.1051/m2an/2023028)
26. L. C. Berselli, A. Kaltenbach, R. Lewandowski, and M. Růžička. *On the existence of weak solutions for a family of unsteady rotational Smagorinsky models*, **Pure and Applied Functional Analysis**, 2023.
URL: <http://yokohamapublishers.jp/online2/oppafa/vol8/p83.html>
27. A. Kaltenbach and M. Růžička. *Existence of steady solutions for a model for micropolar electrorheological fluid flows with not globally log–Hölder continuous shear exponent*, **Journal of Mathematical Fluid Mechanics**, 2023.
DOI: [10.1007/s00021-023-00782-y](https://doi.org/10.1007/s00021-023-00782-y)
28. A. Kaltenbach and M. Růžička. *Analysis of a fully-discrete, non-conforming approximation of evolution equations and applications*, **Mathematical Models and Methods in Applied Sciences**, 2023.
DOI: [10.1142/S0218202523500197](https://doi.org/10.1142/S0218202523500197)
29. S. Bartels and A. Kaltenbach. *Explicit and efficient error estimation for convex minimization problems*, **Mathematics of Computation**, 2023.
DOI: [10.1090/mcom/3821](https://doi.org/10.1090/mcom/3821)

30. A. Kaltenbach and M. Růžička. *Existence of steady solutions for a general model for micropolar electrorheological fluid flows*, **SIAM Journal on Mathematical Analysis**, 2023.
DOI: [10.1137/22M1500599](https://doi.org/10.1137/22M1500599)
31. S. Bartels and A. Kaltenbach. *Error estimates for total-variation regularized minimization problems with singular solutions*, **Numerische Mathematik**, 2022.
DOI: [10.1007/s00211-022-01324-w](https://doi.org/10.1007/s00211-022-01324-w)
32. A. Kaltenbach. *Note on the existence theory for non-induced evolution problems*, **Mathematische Nachrichten**, 2022.
DOI: [10.1002/mana.201900555](https://doi.org/10.1002/mana.201900555)
33. L. C. Berselli, A. Kaltenbach, and M. Růžička. *Analysis of fully discrete, quasi non-conforming approximation of evolution equations and applications*, **Mathematical Models and Methods in Applied Sciences**, 2021.
DOI: [10.1142/S0218202521500494](https://doi.org/10.1142/S0218202521500494)
34. A. Kaltenbach and M. Růžička. *Variable exponent Bochner–Lebesgue spaces with symmetric gradient structure*, **Journal of Mathematical Analysis and Applications**, 2021.
DOI: [10.1016/j.jmaa.2021.125355](https://doi.org/10.1016/j.jmaa.2021.125355)
35. A. Kaltenbach and M. Růžička. *Note on the existence theory for pseudo-monotone evolution problems*, **Journal of Evolution Equations**, 2020.
DOI: [10.1007/s00028-020-00577-y](https://doi.org/10.1007/s00028-020-00577-y)

Review articles

36. S. Bartels and A. Kaltenbach. *Exact a posteriori error control for variational problems via convex duality and explicit flux reconstruction*, **Advances in Applied Mechanics**, 2024.
DOI: [10.1016/bs.aams.2024.04.001](https://doi.org/10.1016/bs.aams.2024.04.001)

Proceedings

37. A. Kaltenbach and M. Růžička. *Conditional quasi-optimal error estimate for a finite element discretization of the p -Navier–Stokes equations: the case $p > 2$* , **Proceedings of the conference 'Critical Phenomena in Nonlinear Partial Differential Equations, Harmonic Analysis, and Functional Inequalities'**, accepted, 2024.
URL: <https://arxiv.org/abs/2411.00043>

Preprints

37. S. Bartels and A. Kaltenbach. *A prox-Based Semi-Smooth Newton Method for Convex Variational Problems*, submitted, 2026.
URL: <https://arxiv.org/abs/2606.25948>
38. H. Antil, A. Kaltenbach, and K. Kirk. *A Finite Element Approximation of an Optimal Insulation Problem with Convective Heat Transfer*, submitted, 2026.
URL: <https://arxiv.org/abs/2606.14991>
39. S. Bartels and A. Kaltenbach. *A prox-Based Semi-Smooth Newton Method for TV-Minimization*, submitted, 2026.
URL: <https://arxiv.org/abs/2605.22728>
40. P. A. Gazca–Orozco and A. Kaltenbach. *A priori and a posteriori error identities for vectorial problems via convex duality*, submitted, 2026.
URL: <https://arxiv.org/abs/2602.04368>
41. H. Antil, A. Kaltenbach, and K. Kirk. *Convection Effects and Optimal Insulation: Modelling and Analysis*, submitted, 2025.
URL: <https://arxiv.org/abs/2512.13098>
42. L. C. Berselli, A. Kaltenbach, and S. Ko. *Error analysis for a fully-discrete finite element approximation of the unsteady $p(\cdot, \cdot)$ -Stokes equations*, submitted, 2025.
URL: <https://arxiv.org/abs/2501.00849>

43. J. Jeßberger and A. Kaltenbach. *Finite element discretization of the steady, generalized Navier–Stokes equations with small shear exponent*, submitted, 2024.
URL: <https://arxiv.org/abs/2408.15731>